```
// C++ code
//
const int lr=10;
const int ly=11;
const int lg=9;
const int trig=6;
const int echo=5;
const int p=A0;
long distance, duration;
void setup()
{
  pinMode(lr, OUTPUT);
  pinMode(ly, OUTPUT);
  pinMode(lg, OUTPUT);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  pinMode(p, INPUT);
  Serial.begin(9600);
}

void loop()
{
  int input=analogRead(p);
  int input1=map(input,0,1023,5,30);

  digitalWrite(trig, LOW);
  delayMicroseconds(2);
  digitalWrite(trig,HIGH);
  delayMicroseconds(10);
  digitalWrite(trig,LOW);

  duration=pulseIn(echo,HIGH);
  distance= 0.034*duration/2;

  if(distance>(input1+30))
  {
    digitalWrite(lr,LOW);
    digitalWrite(ly,LOW);
    digitalWrite(lg,LOW);
    Serial.print("Measured: ");
    Serial.print(distance);
    Serial.print("| Stop Limit: ");
    Serial.print(input1);
    Serial.println("| Zone: Safe");
```

```
  }
  else if(distance>=(input+30) && distance<=(input+15))
  {
    digitalWrite(lr,LOW);
    digitalWrite(ly,LOW);
    digitalWrite(lg,HIGH);
    Serial.print("Measured: ");
    Serial.print(distance);
    Serial.print("| Stop Limit: ");
    Serial.print(input1);
    Serial.println("| Zone: Approach");
  }
  else if(input1<distance<=(input1+15))
  {
    digitalWrite(lr,LOW);
    digitalWrite(ly,HIGH);
    digitalWrite(lg,HIGH);
    Serial.print("Measured: ");
    Serial.print(distance);
    Serial.print("| Stop Limit: ");
    Serial.print(input1);
    Serial.println("| Zone: Caution");
  }
  else
  {
    digitalWrite(lr,HIGH);
    digitalWrite(ly,HIGH);
    digitalWrite(lg,HIGH);
    Serial.print("Measured: ");
    Serial.print(distance);
    Serial.print("| Stop Limit: ");
    Serial.print(input1);
    Serial.print("| Zone: Stop");
  }
}
```